Experiment: 8

AIM: Implement the Blockchain platform Ganache

Theory:

1. Introduction to Ganache

Ganache is a personal blockchain used for Ethereum development and testing. It allows developers to simulate a real blockchain environment locally without any cost or risk.

Ganache is part of the Truffle Suite and is widely used for building, testing, and debugging decentralized applications (DApps).

2. Features of Ganache

- **Instant Block Mining**
  Transactions are processed immediately without delays.
- **Pre-funded Accounts**
  Provides multiple test accounts with virtual Ether.
- **Transaction Monitoring**
  Displays complete transaction history and details.
- **User-Friendly Interface**
  Graphical dashboard to observe blockchain activity such as blocks, gas usage, and balances. ● **Deterministic Environment**
  Ensures consistent results during testing and development.

3. Integration with Development Tools

a) MetaMask

MetaMask acts as a wallet and bridge to connect Ganache with applications.

b) Remix IDE

Remix IDE is used to write, compile, deploy, and interact with smart contracts.

Ganache, MetaMask, and Remix together create a complete local development environment for Ethereum-based applications.

4. Steps to Connect Ganache with MetaMask and Remix IDE

Step 1: Install and Run Ganache

- Download and install Ganache from the Truffle Suite website.
- Launch the application to start a local blockchain.
- It automatically generates accounts with test Ether.

Step 2: Connect Ganache to MetaMask

- Open MetaMask and select **Custom RPC Network**.
- Enter Ganache RPC URL: http://127.0.0.1:7545
- Import Ganache accounts using their private keys.

Step 3: Connect MetaMask with Remix IDE

- Open Remix IDE in a browser.
- Go to **Deploy & Run Transactions** tab.
- Select **Injected Web3** as the environment.
- Remix will connect automatically to the MetaMask account linked with Ganache.

5. Creating a Smart

Contract Example Solidity

Contract pragma solidity

^0.8.0; contract SimpleStorage

{ uint public data; function

set(uint x) public { data = x;

}

}

This contract allows storing and updating a numeric value on the blockchain.

6. Compiling and Deploying the Contract

- Compile the contract using Remix.
- Deploy it using the connected MetaMask account.
- Approve the transaction in MetaMask.

- The contract is deployed on the Ganache local blockchain.

7. Monitoring Transactions in Ganache

- View transaction details such as:
  - Transaction hash
  - Gas usage
  - Block number
- Observe account balance changes in the Ganache interface.

8. Interacting with the Smart Contract

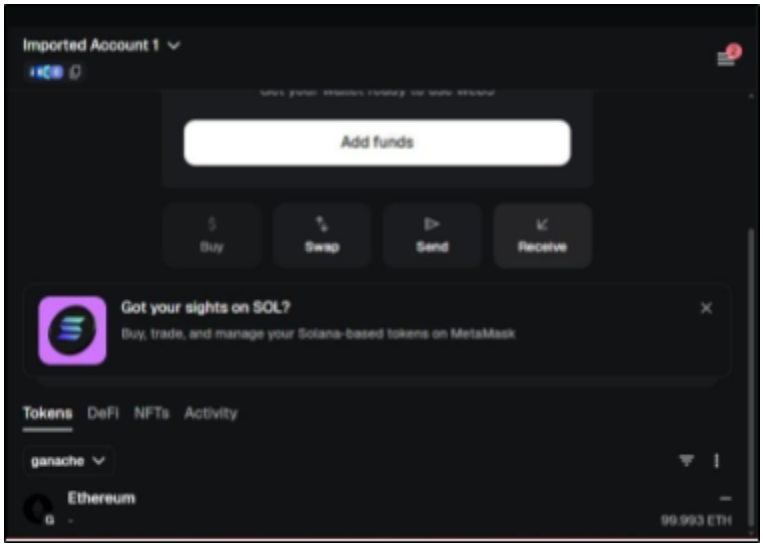
- Use Remix interface to:
  - Call set() to update data (requires transaction) ○ Read stored data using public variable (no gas required) ● Track all state changes and transactions in Ganache.

Advantages of Using Ganache

- No real cryptocurrency required
- Fast and efficient testing
- Easy debugging of smart contracts
- Complete control over blockchain environment

CODE:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.19;

contract PersonalHealthRecord {

   struct Record { string ipfsHash;
string recordType; uint256
timestamp; address addedBy;
   }

   struct Patient { string name;
uint age; bool exists;
   }

   mapping(address => Patient) public patients;
mapping(address => Record[]) private records;
mapping(address => mapping(address => bool))
public accessPermission;

   event PatientRegistered(address patient); event
RecordAdded(address patient, string ipfsHash); event
AccessGranted(address patient, address doctor);
event AccessRevoked(address
patient, address doctor);

   modifier onlyPatient() {
require(patients[msg.sender].exists, "Patient
not registered");
      _;
   }

   function registerPatient(string
   memory
_name, uint _age) public {
require(!patients[msg.sender]
.exists, "Already
registered");

      patients[msg.sender] =
      Patient(_name,
_age, true);

      emit
      PatientRegistered(msg.sende
      r);
   }
```



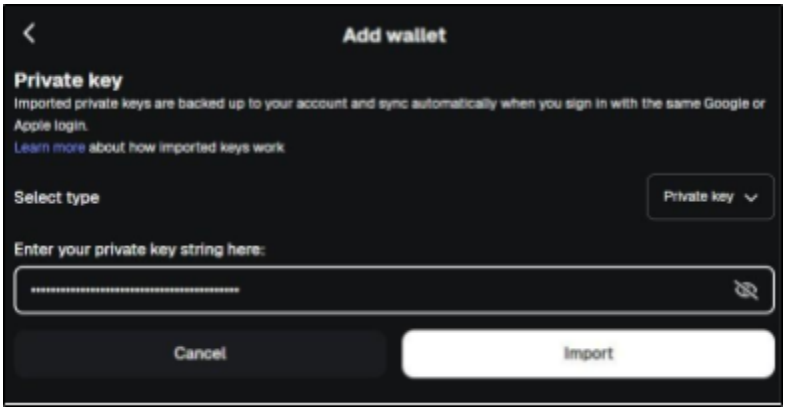

```
function addRecord(        string
memory _ipfsHash,        string
memory _recordType
) public onlyPatient {

    records[msg.sender].push(
Record({                ipfsHash: _ipfsHash,
recordType: _recordType,
timestamp: block.timestamp,
addedBy: msg.sender
        })
    );

    emit RecordAdded(msg.sender, _ipfsHash);
}
```

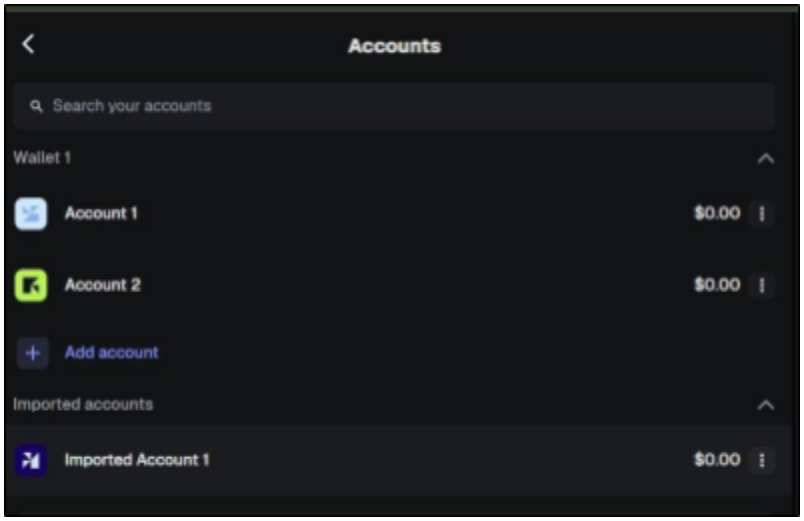
OUTPUT : Metamask:

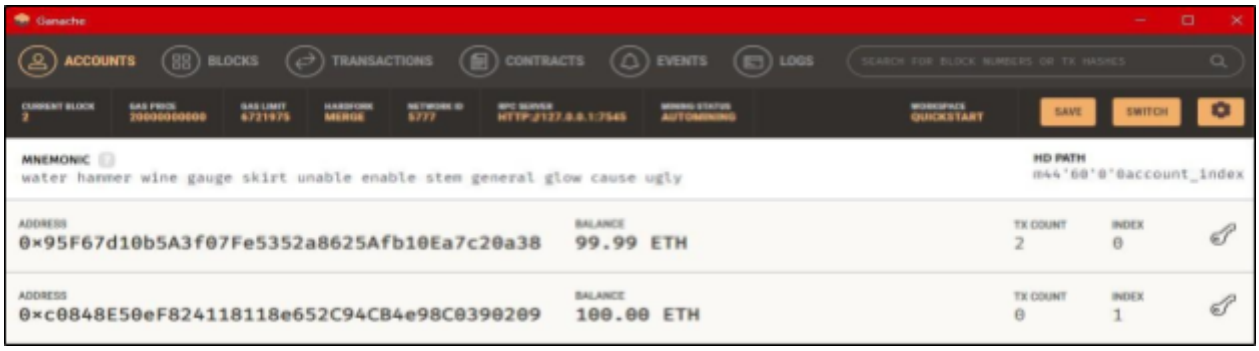
```
function
grantAccess(addres
s doctor) public
onlyPatient {
accessPermission[msg.sender

}
][doctor] = true;        emit
                         }
```

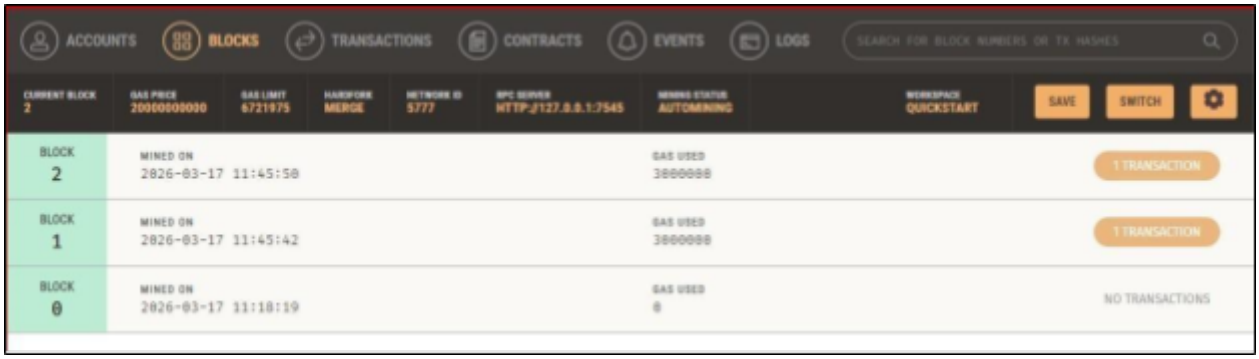
Added the wallet by giving the private key of my wallet:

```
AccessGranted(msg.se
nder, doctor);
}

function revokeAccess(address doctor) public
onlyPatient {
accessPermission[msg.sender][doctor] = false;
emit AccessRevoked(msg.sender, doctor);
}

function getRecords(address patient)
public view returns (Record[] memory)
{        require(        patient ==
msg.sender ||
accessPermission[patient][msg.sender]
,
        "Access denied"
);

return records[patient];
```


Add wallet
Private key
Imported private keys are backed up to your account and sync automatically when you sign in with the same Google or Apple login.
Learn more about how imported keys work
Select type
Private key
Enter your private key string here:
Cancel
Import


Accounts
Search your accounts
Wallet 1
Account 1
$0.00
Account 2
$0.00
Add account
Imported accounts
Imported Account 1
$0.00


Ganache
ACCOUNTS
BLOCKS
TRANSACTIONS
CONTRACTS
EVENTS
LOGS
SEARCH FOR BLOCK NUMBERS OR TX HASHES
CURRENT BLOCK 2
GAS PRICE 20000000000
GAS LIMIT 6721975
HARDFORK MERGE
NETWORK ID 5777
RPC SERVER HTTP://127.0.0.1:7545
MINING STATUS AUTOMINING
WORKSPACE QUICKSTART
SAVE
SWITCH
MNEMONIC
water hammer wine gauge skirt unable enable stem general glow cause ugly
HD PATH
m/44'/60'/0'/0/account_index
ADDRESS
0x95F67d10b5A3f07Fe5352a8625Afb10Ea7c20a38
BALANCE
99.99 ETH
TX COUNT
2
INDEX
0
ADDRESS
0xc0848E50eF824118118e652C94CB4e98C0390209
BALANCE
100.00 ETH
TX COUNT
0
INDEX
1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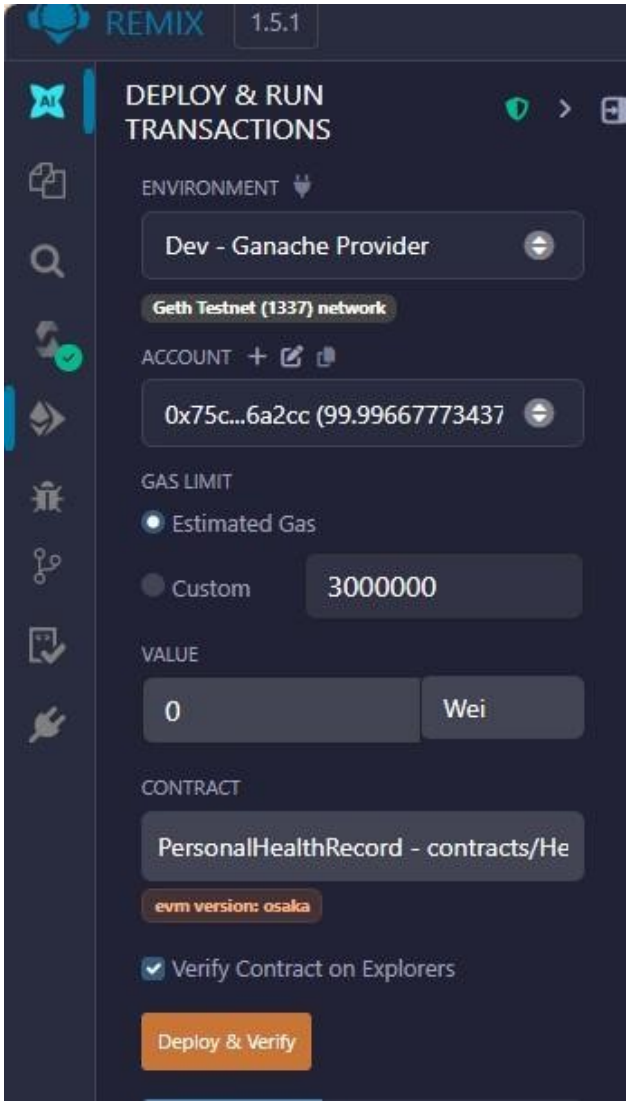
HARDFORK MERGE
NETWORK ID 5777
RPC SERVER HTTP://127.0.0.1:7545
MINING STATUS AUTOMINING
WORKSPACE QUICKSTART
SAVE
SWITCH
BLOCK 2
MINED ON 2026-03-17 11:45:50
GAS USED 3000000
1 TRANSACTION
BLOCK 1
MINED ON 2026-03-17 11:45:42
GAS USED 3000000
1 TRANSACTION
BLOCK 0
MINED ON 2026-03-17 11:18:19
GAS USED 0
NO TRANSACTIONS



Conclusion:

Ganache provides a powerful local blockchain environment for testing smart contracts. When combined with MetaMask and Remix IDE, it enables developers to efficiently build, deploy, and test decentralized applications in a safe and controlled setup.